

NATIONAL INSTITUTE OF TECHNOLOGY DURGAPUR
Even Semester Mid-term Examination, 2025-26

Course Code: CEC 403

Course Name: Water Resource Engineering

Full Marks: 25

Time: 90 Mins

Instructions: Answer all the questions.
 Materials to be supplied: None

Question No.	Body of the Question	Marks	Mapped CO
1	(a) Draw a flood hydrograph for a watershed before and after urbanization and indicate one major difference.	1	CO1
	(b) If the rainfall intensity is less than the infiltration capacity of soil, what will be the actual infiltration rate?	1	
	(c) At what depths is velocity measured using a current meter in shallow and deep streams?	1	
	(d) What is a rating curve? Draw a typical sketch.	1	
	(e) In the SCS Curve Number method, what do CN = 100 and CN = 0 represent?	1	
	(f) Name the type of rain gauges that directly provides a mass curve of rainfall.	1	
2	Determine the drainage area at the outlet of the catchment which has a triangular unit hydrograph having time based of 16 hours and peak discharge of 28 m ³ /s.	2	CO2
3	The rainfall during three successive 2 h periods are 0.5, 2.8 and 1.6 cm. The surface runoff resulting from this storm is 3.2 cm. What is the ϕ (phi) index value of this storm?	3	CO1
4	There are four rain gauge stations existing in the catchment. The average annual rainfall values at these stations are 750, 600, 420 & 500 mm respectively. Determine the optimum number of a rain gauges in the catchment if it is desired to limit the error in the mean value of a rainfall in the catchment to 10%. Is the number of rain-gauge in that catchment sufficient to record the amount of rainfall? If not, how many more gauges are required to be installed?	3	CO2
5	A catchment is idealized as a 25 km × 25 km square. It has five rain gauges, one at each corner and one at the center. Gauge location is as follow: G1 – Top left corner, G2 – Top right corner, G3 – Bottom right corner, G4 – Bottom left corner, G5 – Center of the catchment. During a month, the precipitation at these Gauges is measured as G1= 300mm, G2= 285mm, G3=272mm, G4=290 mm and G5= 288 mm. Calculate the average precipitation (in mm up to one decimal place) over a catchment during this month using Thiessen Polygon Method.	4	CO1
6	The isohyets for annual rainfall over a catchment basin indicate the following (isohyet, area): (45-55cm, 500 km ²); (55-65, 1000 km ²) and (65-75, 2000 km ²). What is the approximate average annual precipitation depth (in cm) over the catchment area?	3	CO1
7	The ordinates of a 2-hour Unit Hydrograph of a catchment are 0 m ³ /s at 0 hr, 12 m ³ /s at 2 hr, 30 m ³ /s at 4 hr, 24 m ³ /s at 6 hr, 14 m ³ /s at 8 hr, 6 m ³ /s at 10 hr, and 0 m ³ /s at 12 hr. Convert this 2-hour Unit Hydrograph into a 6-hour Unit Hydrograph using the principle of superposition.	4	CO2

Course Outcomes

CO1: Understanding of occurrence, distribution, storage & transmission of water in different form in the space, on the surface and below the surface of the earth

CO2: Understanding of tempo-spatial collection of data and preparation of hydrometeorological information system

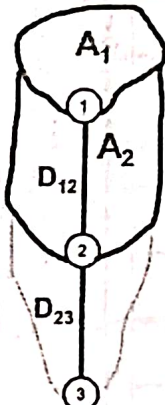
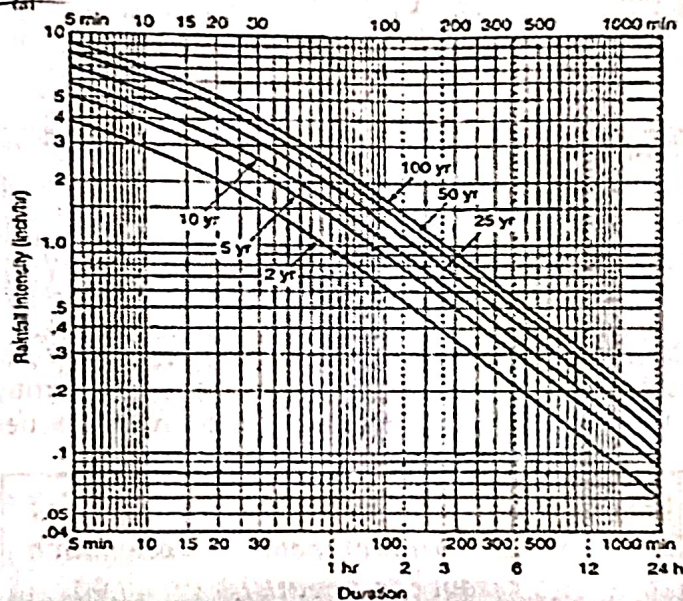
CO3: learning importance, requirement, method & infrastructure for imparting irrigation water to crop, development & conservation of water for its economic & efficient use

$\frac{4}{6} = \frac{2}{3}$

$(5) = 1 + \frac{2}{2} \quad (\frac{2}{3})$

$3 + (2) = 3 + \frac{2}{2} = 3 + 1 = 4$

Instructions: Answer all the questions.
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Question No.	Body of the Question	Marks	Mapped CO																				
1	Design the diameters of storm sewer pipes (D_{12} and D_{23}) for a 2-year return period storm for an urban catchment. Each pipe has a length of 500 m and Manning's roughness coefficient $n = 0.014$. The relevant catchment characteristics are provided in the table. The rainfall intensity-duration-frequency (IDF) relationship is given in inches per hour; convert it into SI units (mm/hr) using the relation 1 inch = 25.4 mm. <table><thead><tr><th>Catchment No</th><th>Elevation at the inlet (m)</th><th>Area of watershed (Km²)</th><th>Coefficient (C)</th><th>Time of overland flow (t_o)</th></tr></thead><tbody><tr><td>1</td><td>100</td><td>0.005</td><td>0.5</td><td>10</td></tr><tr><td>2</td><td>97</td><td>0.006</td><td>0.6</td><td>5</td></tr><tr><td>3</td><td>95</td><td>-</td><td>-</td><td>-</td></tr></tbody></table>	Catchment No	Elevation at the inlet (m)	Area of watershed (Km ²)	Coefficient (C)	Time of overland flow (t _o)	1	100	0.005	0.5	10	2	97	0.006	0.6	5	3	95	-	-	-	10	CO1
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2	97	0.006	0.6	5																			
3	95	-	-	-																			
2	Explain the main components of a diversion headwork with a neat sketch and describe the function of each component.	5	CO3																				
3	Differentiate the following: i) Borrow pit vs spoil bank ii) Hydrograph vs hyetograph iii) Duty Vs Delta iv) Direct Runoff Hydrograph (DRH) vs Flood Hydrograph v) Base period vs crop period	5	CO1																				

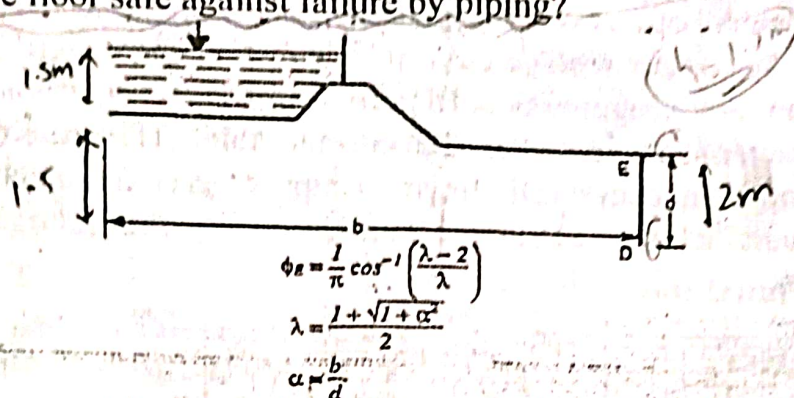
Course Outcomes

CO1: Understanding of occurrence, distribution, storage & transmission of water in different form in the space, on the surface and below the surface of the earth

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The concrete floor of a head regulator is level with channel bed and is 13 m long. The floor is provided with cut off wall at its upstream and downstream ends. The depth of u/s cutoff is 1.5 m (below the floor level) and that of d/s wall is 2.0 m. Using Khosla's theory (see figure below for sketch and formula), determine the thickness at its mid length and also at its junction with the u/s and d/s cutoff wall. The FSL is 1.5 m above the floor level. if the permissible exit gradient is 0.18, is the floor safe against failure by piping?



10

CO3

Convert the following 2-hr Unit hydrograph into 3-hr Unit Hydrograph using S-curve method.

Time (hr)	0	1	2	3	4	5	6	7	8	9	10
Q (m ³ /s)	0	75	250	300	275	200	100	75	50	25	0

10

CO2

Determine the discharge in the stream for the given data at a gauging station below. the rating equation of current meter may be taken as $V = 0.45N_s + 0.025 \text{ m/s}$, where, N_s is the revolution per second.

Distance from left	Depth (m)	Revolution of a current meter kept at a depth of 0.6 m	Duration of observation (s)
0	0	0	0
1.1	1.2	41	110
3.2	2.7	62	110
5.3	2.7	115	160
7.1	2.3	95	153
9.2	1.9	47	105
11.6	1.7	35	107
12.5	0	0	0

10

CO2

Describe and sketch any five types of river training works?

5

CO1

What is meant by cross drainage work? Name the different types of cross drainage work with sketch.

5

CO3